LEARN CODING

ale66

github.com/ale66/learn-coding

MODELS OF COMPUTATION

A general approach to agency [Simon, McCarthy]

To act intelligently in its context, an [AI] agent needs to keep an internal representation of the world around by which it understands sensory inputs

see code and understand what it does *by running it in your mind*

Reason about a task, visualise a code in your mind, then write it (on VS Code) then run it. And it does.

mental models must have a reasonable representation of computer's inner workings

MENTAL MODELS OF COMPUTERS

- Computers are complex devices
- almost no one fully understands their workings all the way down to electronics.
- we need abstractions
  - inspired by human pen-and-paper computation
  - support code comprehension

FOUR MODELS OF COMPUTATION

1. human operations at the blackboard, with parentheses, etc.
2. Spreadsheet operations: each cell computes a simple expression
3. Central memory operations: traditional computation with program counter, iterations etc.
4. Vectorial computation: linear algebra on GPU hardware

EXPERIENCE: BLACKBOARD COMPUTATION

- instructor write all data on board then rewrites it step by step
- simple operations take place in your mind, longer operations are written down in a step by step fashion
- only one person operates, he/she writes top-down, left-right

EXPERIENCE: SPREADSHEET COMPUTATION

- each cell has either a fixed input data or a formula
- one formula cell can depend on many other cells, via aggregation/iteration (compute average, max, min etc.)
- when an input data cell changes, all formula cells are re-calculated
- any cell can use any other cell, even though we tend to arrange top-down, left-right computation
- no circular definitions please (`A1 = B1*1.22` with `B1=A1*0.82` 💣)
- funny `=` which actually means ←

EXPERIENCE: CENTRAL MEMORY COMPUTATION

- no fixed cells, memory locations are introduced as we go along
- a hidden marker, the *program counter* points to the next operation to be executed
- visualise it on pythontutor.com/ or Memory graph memory-graph.com/

EXPERIENCE: VECTORIAL COMPUTATION

- the basis of modern computation (AV streaming, Video Games, AI)
- implements vector/matrix algebra (instead of simple scalar algebra from school)
- will see it, if time allows, at the end